Q1.

A child, of mass 40 kg, moves at constant speed of 5.0 m s^{-1} on a fairground ride.

The path of the child is a circle of radius 4.0 metres.

Find the magnitude of the resultant force acting on the child.

Circle your answer.

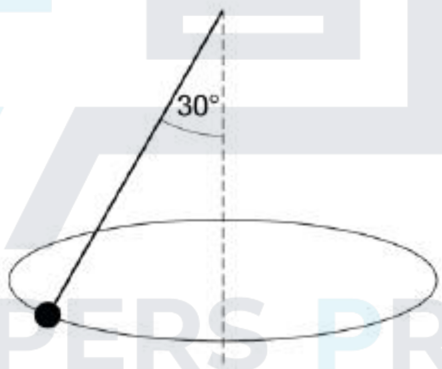
6.3 N   50 N   130 N   250 N

(Total 1 mark)

Q2.

In this question use *g* = 9.8 m s^{-2}.

A conical pendulum consists of a string of length 60 cm and a particle of mass 400 g. The string is at an angle of 30° to the vertical, as shown in the diagram.



(a) Show that the tension in the string is 4.5 newtons.

(2)

(b) Find the angular speed of the particle.

(3)

(c) State **two** assumptions that you have made about the string.

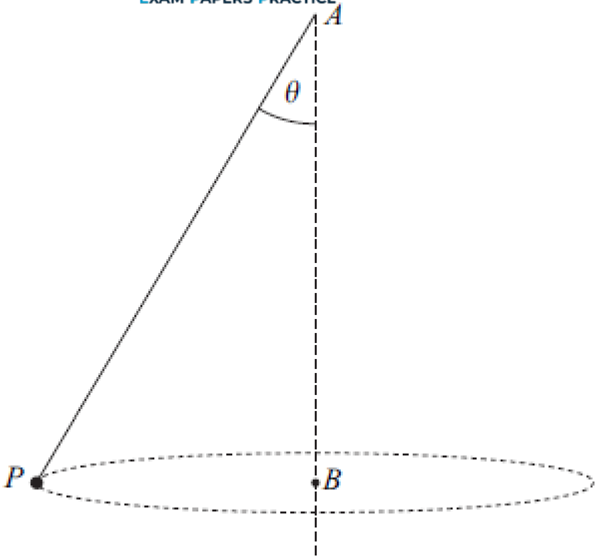
(1)

(Total 6 marks)

Q3.

A light inextensible string has one end attached to a particle, P, of mass 2 kg. The other end of the string is attached to the fixed point A. The point A is vertically above the point B. The particle moves at a constant speed in a horizontal circle of radius 0.8 m and centre B. The tension in the string is 34 N.

The string is inclined at an angle θ to the vertical, as shown in the diagram.



(a) Find the angle θ.

(3)

(b) Find the speed of the particle.

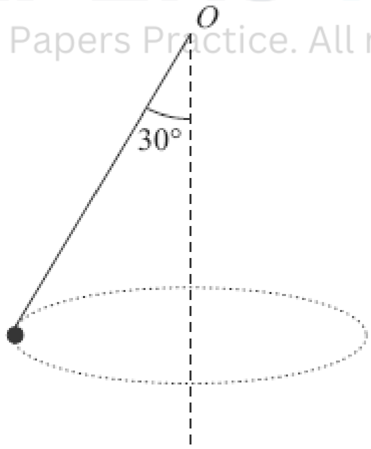
(3)

(c) Find the time taken for the particle to make one complete revolution.

(2)
(Total 8 marks)

Q4.

A particle, of mass 6 kg, is attached to one end of a light inextensible string. The other end of the string is attached to the fixed point O. The particle is set in motion, so that it moves in a horizontal circle at constant speed, with the string at an angle of 30° to the vertical. The centre of this circle is vertically below O.



The particle moves in a horizontal circle with an angular speed of 40 revolutions per minute.

(a) Show that the angular speed of the particle is $\frac{4\pi}{3}$ radians per second.

(2)

(b) Show that the tension in the string is 67.9 N, correct to three significant figures.

(3)

(c) Find the radius of the horizontal circle.

(4)

(Total 9 marks)

Q5.

A car of mass 1200 kg travels round a roundabout on a horizontal, circular path at a constant speed of 14 m s^{-1}. The radius of the circle is 50 metres. Assume that there is no resistance to the motion of the car and that the car can be modelled as a particle.

(a) A friction force, directed towards the centre of the roundabout, acts on the car as it moves. Show that the magnitude of this friction force is 4704 N.

(4)

(b) The coefficient of friction between the car and the road is μ. Show that $\mu \geq 0.4$.

(3)

(Total 7 marks)

Q6.

A motorcyclist rides her motorcycle at a constant speed on a horizontal circular path around a roundabout.

(a) The radius of the circular path of the motorcycle is 15 metres.

(i) Calculate the magnitude of the acceleration of the motorcycle when it is moving at 7.5 m s^{-1}.

(2)

(ii) The total mass of the motorcycle and rider is 400 kg. To avoid skidding taking place, the magnitude of the frictional force between the motorcycle and the road must not exceed 2940 N. Find the greatest speed, V m s^{-1}, at which the motorcyclist can ride her motorcycle around this roundabout without skidding.

(3)

(b) Without referring to air resistance, state one modelling assumption you have made in answering part (a). Explain why your assumption is reasonable.

(2)

(c) In wet weather the maximum frictional force between the motorcycle and the road is reduced. The motorcyclist wishes to ride her motorcycle at the speed V m s^{-1}, as found in part (a)(ii), around the roundabout. State whether she should increase or decrease the radius of her circular path to avoid skidding, giving a reason for your answer.

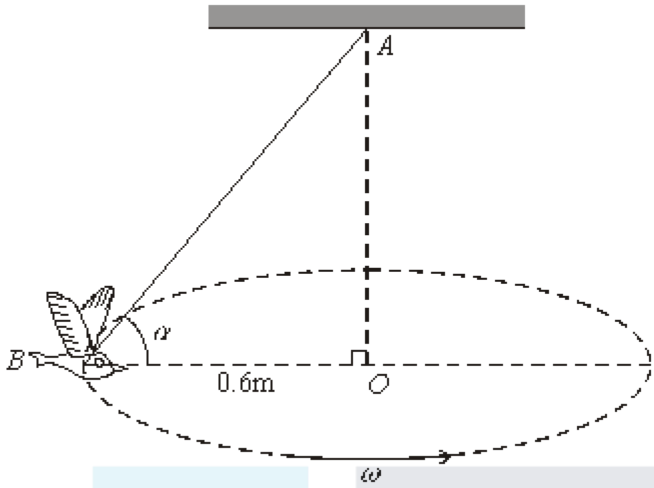
(2)

(Total 9 marks)

Q7.

A toy bird, of mass 0.25 kg, is fixed to one end, B, of a light inextensible string. The other end, A, of the string is attached to a fixed point on a ceiling. The bird is set in motion, so that it describes a horizontal circle of radius 0.6 m. The centre of this circle is O, which is vertically below A, as shown in the diagram. The angle ABO is α.

The bird moves with a constant angular speed of ω radians per second.



The bird completes one full revolution every 1.5 seconds.

(a) Find ω, leaving your answer in terms of π.

(2)

(b) (i) Determine the magnitude of the acceleration of the bird.

(2)

(ii) Show the direction of this acceleration on a diagram.

(1)

(c) (i) Draw a diagram to show the forces acting on the bird.

(1)

(ii) Find the angle α, giving your answer to the nearest degree.

(6)

(d) State **one** modelling assumption used in this question.

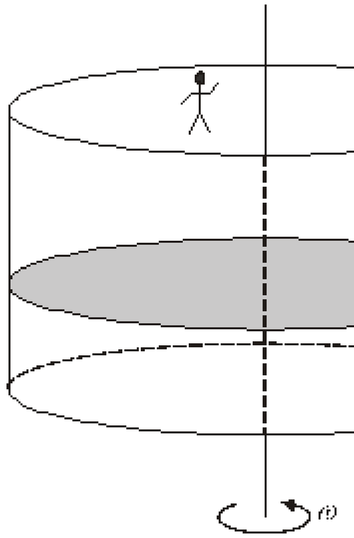
(1)

(Total 13 marks)

Q8.

A ride in an amusement park consists of a hollow cylindrical cage of radius 3 m, with a horizontal floor and vertical sides.

Martin, who has a mass of 40 kg, stands at the edge of the floor of the cage, against the wall. The cage begins to rotate about its vertical axis. When the cage is rotating at a constant angular speed, ω radians per second, the cage floor is lowered. Martin remains in the position shown in the diagram due to the vertical frictional force between him and the wall.



(a) Draw a diagram showing the forces acting on Martin after the floor has been lowered.

(1)

(b) (i) State the magnitude of the frictional force.

(1)

(ii) The magnitude of the normal reaction between Martin and the wall is 784 N. Find the least possible value of the coefficient of friction between Martin and the wall.

(2)

(iii) Show that ω = 2.56, correct to three significant figures.

(3)

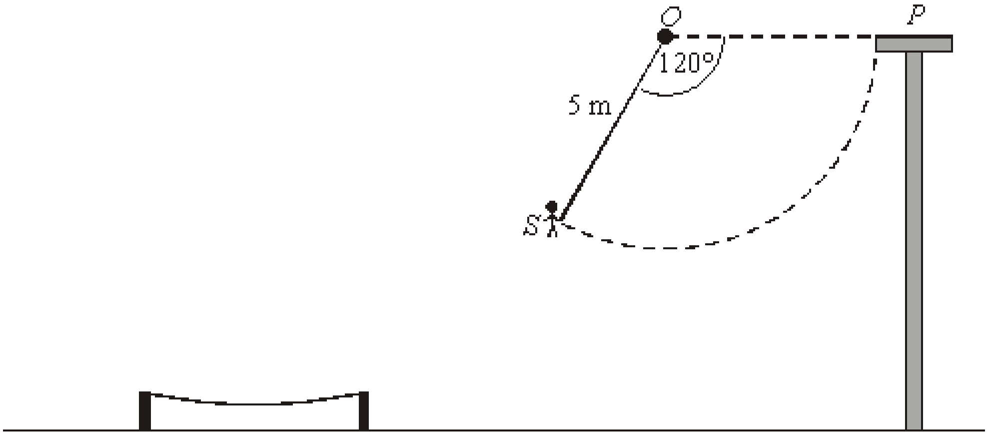
(c) State **one** modelling assumption used in this question.

(1)

(Total 8 marks)

Q9.

A stuntman S stands initially on a platform P. He holds a rope which is attached to a fixed support O. The points O and P are at the same horizontal level. The man then steps off the platform with the rope taut and horizontal. He swings through a circular arc, as shown in the diagram below.



The mass of the man is 70 kg and the length of the rope is 5 m. The man may be modelled as a particle throughout the motion.

(a) The man releases his grip on the rope when it has swung through an angle of 120°.

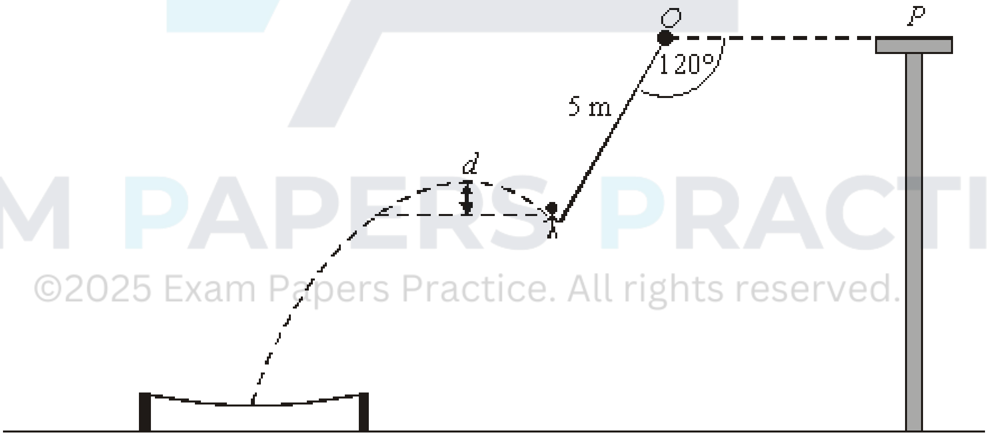
(i) Show that the speed of the man when he is about to let go of the rope is approximately 9.21 m s^{-1}.

(4)

(ii) Find the tension in the rope when the man is about to let go of the rope.

(5)

(b) After letting go of the rope, the stuntman moves freely under gravity.



He rises a distance d before beginning to fall into the safety net. Show that d is approximately 1 metre.

(4)

(c) Comment on the assumption that the man can be modelled as a particle.

(1)
(Total 14 marks)

Q10.

A spin drier is tested when empty. The drum, initially at rest, reaches its maximum angular speed of 1 100 revolutions per minute in 8 seconds.

(a) Express the maximum angular speed of the drum in radians per second.

(2)

(b) Assuming that the angular acceleration of the drum is constant, calculate its magnitude.

(2)
(Total 4 marks)

Q11.

The orbit of the moon around the earth may be modelled as circular. The time taken for the moon to make one complete orbit is approximately 27.32 days.

(a) Show that the angular speed of the moon is approximately 2.66×10^{-6} radians per second.

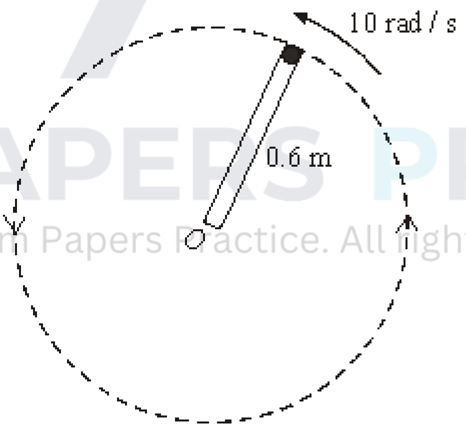
(3)

(b) Assuming that the radius of the circular orbit is approximately 3.844×10^8 metres, find the speed of the moon, relative to the earth, in metres per second.

(2)
(Total 5 marks)

Q12.

The diagram shows a smooth tube, sealed at each end, which rotates about one end. There is a small sphere inside the tube. The tube rotates in a vertical plane, so that the sphere remains at the end of the tube and describes a vertical circle of radius 0.6 metres. The mass of the sphere is 0.05 kg and the tube rotates with a constant angular speed of 10 radians per second.



(a) Calculate the magnitude of the acceleration of the sphere.

(2)

(b) Find the magnitude of the horizontal reaction force exerted by the end of the tube on the sphere when the tube is horizontal.

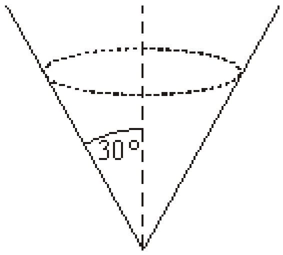
(2)

(c) Find the magnitude of the vertical reaction force exerted by the end of the tube on the sphere when the sphere is at its lowest point.

(3)
(Total 7 marks)

Q13.

A particle, of mass 3 kg, describes a horizontal circular path on the inside surface of a smooth cone, as shown in the diagram.



The radius of the circle is 0.5 metres and the semi-vertical angle of the cone is 30°. The particle moves at a constant speed.

(a) (i) Show that the magnitude of the normal reaction force on the particle is 58.8 N.

(3)

(ii) Find the speed of the particle.

(4)

(b) The particle moves on the same cone, but in a horizontal circle of greater radius than before.

(i) What happens to the magnitude of the normal reaction force?

(1)

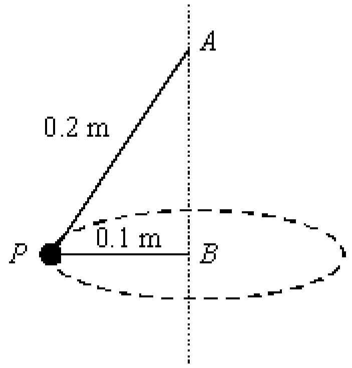
(ii) What happens to the speed of the particle? Explain your answer.

(2)

(Total 10 marks)

Q14.

A particle, P, of mass 2 kg, is attached to two strings of lengths 0.2 metres and 0.1 metres. The strings are fixed to the particle and to the points A and B respectively. The point A is directly above the point B. The particle describes a horizontal circle, centre B and radius 0.1 metres, at a speed of 4 m s^{-1} The particle and strings are shown in the diagram.



(a) Calculate the magnitude of the acceleration of the particle.

(2)

(b) Find the tension in the upper string.

(4)

(c) Find the tension in the lower string

(5)
(Total 11 marks)